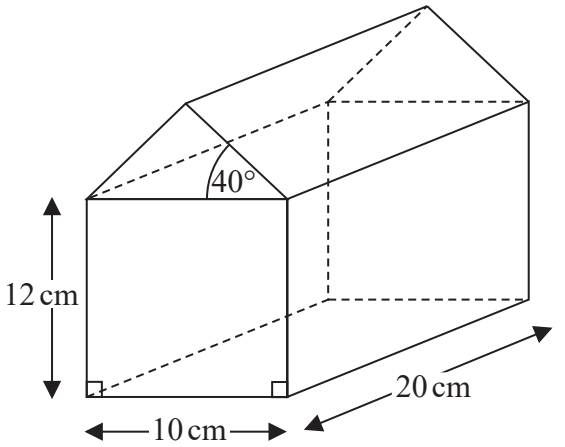
9 The diagram shows a prism.



The cross section of the prism has exactly one line of symmetry.

Work out the volume of the prism.
Give your answer correct to 3 significant figures.

Symmetric roof: half-width = 10/2 = 5 cm.

Roof height = 5 tan 40° ≈ 4.1955 cm.

Cross-section area = 10 × 12 + ½ × 10 × 4.1955...

≈ 140.9775 cm^2. Volume = area × 20

≈ 2819.55... cm^3 = 2820 cm^3 (3 s.f.).

5 marks total
Half-width 5
Height 5tan40°
Rectangle + triangle
Multiply by 20
2820 cm^3

..... cm^3

(Total for Question 9 is 5 marks)

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